MOHAN SATHWIK GUDLA

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OBJECTIVE

Data-driven professional with 2+ years of experience building analytical models and decision-support systems to improve operational efficiency, cost optimization, and product quality. Skilled in Python, machine learning, statistical modeling, and data visualization to extract actionable insights from large, complex datasets. Experienced in developing optimization models and predictive analytics solutions that support strategic business decisions. Seeking to leverage data science and analytics expertise to solve real-world business problems and drive measurable impact.

EDUCATION

Bachelor of Technology, VIT University - Vellore Campus — 2019 - 2023

Relevant Coursework: Software Engineering, Statistics, Data Analytics, Data Science and Machine Learning.

SKILLS

Languages & Frameworks	Python (Pandas, NumPy, Matplotlib, Scikit learn), SQL
Databases & Platforms	MySQL, SSMS, Databricks, Cosmos DB
BI & Reporting Tools	Power BI, Excel
Data Analysis & Statistical Skills	Statistical Modeling (Regression, Forecasting, Variance Analysis), Time Series Analysis, Hypothesis Testing
Soft Skills	Logical Thinking and Problem Solving, Good with Communication and Collaboration.
Others	Azure, Power Automate

EXPERIENCE

Data Science Analyst — Aug 2023 - present

PepsiCo GBS — *Hyderabad, India*

- Built data-driven decision support systems using Python and statistical modeling to improve operational planning, supplier selection, and product quality optimization across multiple regions.
- Analyzed large operational datasets covering supplier performance, product quality metrics, environmental factors, and logistics variables to identify key performance drivers and improvement opportunities.
- Conducted large-scale exploratory data analysis and feature engineering on **100+** variables, uncovering seasonality patterns and performance variability across suppliers and operational segments.
- Developed clustering-based analytics models (K-Means) to segment suppliers and product variants by performance, enabling data-driven sourcing strategies and improving decision transparency.
- Designed a statistical opportunity detection engine that identifies high-impact optimization windows, revealing improvement opportunities in around **20–50%** of operational time periods.
- Built a linear programming optimization model to recommend optimal supplier allocation plans under multiple constraints such as capacity, demand targets, and supply limitations.
- Identified **12%** potential cost savings in procurement planning by optimizing supplier allocation and reducing inefficient resource distribution.
- Developed predictive models to estimate performance potential at the operational unit level, identifying improvement opportunities of **8–19%** in productivity through optimized operational practices.
- Built interpretable statistical models to analyze non-linear relationships between operational practices and product quality metrics, achieving **50–57%** model explanatory power.
- Reduced manual analysis effort by **60–70%** by automating data pipelines, modeling workflows, and decision-support dashboards, accelerating insight generation and strategic planning.

- Collaborated with solution architects on end-to-end data architecture design and implemented dynamic allocation logic during a global rollout, ensuring data accuracy and seamless transition across systems.
- Performed quantitative variance analysis and root-cause diagnostics on datasets to identify reporting anomalies and control gaps.
- Documented end-to-end reporting workflows, data definitions, and control procedures to ensure audit readiness, governance compliance, and sustainable process ownership.

PROJECTS

Cost Optimization & Scenario Modeling Project. Designed and deployed a scenario based modeling and cost optimization solution integrating predictive quality models with budgetary and operational constraints, enabling quantitative evaluation of cost, risk, and margin trade-offs through interactive BI reporting.

Productivity Prediction & Optimization. Developed predictive models to estimate unit-level productivity potential using operational and environmental datasets. Performed data preprocessing, feature engineering, and clustering to identify key drivers influencing productivity.Identified improvement opportunities of in productivity through optimized operational practices.

Product Quality Optimization using Statistical Modeling Built Generalized Additive Models (GAM) to analyze non-linear relationships between operational practices and product quality metrics.Engineered features from multi-source datasets and applied feature selection to identify key drivers impacting quality outcomes. Improved model explanatory power and delivered simulation dashboards to support data-driven operational decisions.

CERTIFICATIONS

- Azure Fundamentals (AZ-900)
- Azure Data Fundamentals (DP-900)
- Microsoft Power BI Data Analyst (PL-300)
- NASSCOM Big Data Fundamentals

HACKATHONS

AWS x PepsiCo Gen-AI Hackathon (Intelligent Promotion Planner Agent)

- Built an AI-driven promotion planning solution using AWS (S3, Athena, Lambda, SageMaker, Bedrock) to automate and optimize promotional workflows.
- Developed data integration and predictive intelligence components enabling faster planning and improved promotion insights.
- Contributed to a prototype demonstrating 40% faster planning, 50% improved projected ROI, and potential revenue opportunity.